

PROPOSTA DI GRANT GIOVANI IN CSN5 - Anno 2026

Acronimo:	PRISM
Titolo completo:	Precision Room-temperature Instrumentation for high-energy x-ray absorption Spectroscopy Measurements
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Abstract (2000 caratteri)

X-ray absorption spectroscopy (XAS) is a powerful element-selective technique for probing the electronic and atomic structure of materials. While synchrotron and free-electron laser facilities provide outstanding performance, their limited accessibility motivates advanced laboratory instrumentation. A major challenge is extending fluorescence-mode XAS beyond 20 keV, where silicon detectors lose efficiency and germanium detectors require cryogenic operation. This limits studies of elements such as Mo, Rh, Ag and Cd, whose K-edges are relevant to catalysis, electrochemistry, advanced alloys and energy materials. PRISM will develop a high-resolution laboratory hard X-ray spectrometer combining advanced crystal optics with a multichannel cadmium zinc telluride (CZT) detector for precision fluorescence spectroscopy up to 50 keV. Building on my expertise in crystal-based X-ray spectroscopy gained within the VOXES project and SIDDHARTA-2 experiment, PRISM will develop detector architectures, low-noise front-end electronics, calibration procedures and spectral reconstruction algorithms to achieve sub-keV energy resolution, high quantum efficiency (>90% up to ~40 keV) and excellent SNR performance at room temperature. The instrument will be validated through fluorescence XAS measurements on reference materials with K-edges above 20 keV, demonstrating accurate edge reconstruction and XANES spectra within laboratory-compatible acquisition times. The resulting capability will enable extended in situ and operando studies of slowly evolving catalytic, electrochemical and energy-related processes. Beyond this proof of concept, PRISM will establish a new detector platform for next-generation laboratory hard X-ray spectroscopy, strengthening INFN expertise in semiconductor radiation detectors, precision X-ray instrumentation and crystal spectroscopy. The technology will complement future photon-source infrastructures such as EuAPS and support research and industrial applications.

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1 State of the art (4000 caratteri)

A central challenge in materials characterization is determining how electronic and local atomic structure evolve under realistic conditions. Changes induced by irradiation, processing, phase transformations, chemical reactions or device operation may be transient, reversible or environment-dependent, and can therefore be missed by ex-situ measurements [1]. This requires in-situ and/or operando techniques capable of probing thick, heterogeneous or diluted samples while retaining sensitivity to selected elements and their local coordination [2,3]. Hence, there is a broad need for non-destructive, element-selective and bulk-sensitive probes compatible with complex environments and long acquisition campaigns.

X-rays are particularly suited for such investigations because of their high penetration depth with respect to other optical techniques, enabling access to the bulk of samples. By tuning the photon energy of a selected element, X-ray spectroscopies provide element- and orbital-specific information, even in chemically complex and heterogeneous systems. These properties make X-ray methods well-suited to non-destructive in situ and operando studies [4].

Among X-ray techniques, X-ray absorption spectroscopy (XAS) is particularly suited to this purpose because it measures the energy-dependent absorption coefficient across an element-specific absorption edge [5,6]. The near-edge region, the XANES, is sensitive to oxidation state, electronic structure and coordination geometry, whereas the extended region up to 1 keV above the edge, the EXAFS, provides quantitative information on the local atomic environment, including interatomic distances, coordination numbers and structural disorder. XAS can be measured in transmission, by recording the attenuation of the incident beam through the sample, or in fluorescence, by detecting the characteristic X-rays emitted after absorption. While transmission is generally preferred for homogeneous samples of suitable thickness, fluorescence detection is essential for diluted absorbers, thick specimens and complex sample environments, where the transmitted intensity is too weak or difficult to access.

Synchrotron facilities are the reference platforms for XAS, as their high brilliance, broad energy tunability and dedicated beamlines enable high-quality spectra with excellent signal-to-noise ratio over a wide energy range. X-ray free-electron lasers further provide ultrashort and intense X-ray pulses, opening access to femtosecond time-resolved experiments and transient states. However, access to these large-scale facilities is limited and competitive, and the allocated beamtime is often unsuitable for routine characterization, long experimental campaigns, or studies involving hazardous and radioactive samples [7].

These limitations have driven renewed interest in laboratory XAS systems based on conventional X-ray tubes [8]. Although laboratory sources provide substantially lower brilliance than synchrotrons, they offer continuous access, flexible sample handling and long measurement times, making them attractive for routine analysis, method development and slowly evolving in situ or operando processes. Recent advances in source brightness, crystal optics and detection technology have enabled an increasing number of XAS and XES measurements to be transferred back to the laboratory.

Within INFN, VOXES [9] has established expertise in high-resolution von Hamos spectroscopy for extended and dilute X-ray sources. The instrument currently operates mainly in the 5–20 keV range and has recently been upgraded with motorized positioning, energy-dispersive monitoring and mechanical compatibility with laboratory XAS measurements [10].

Despite this progress, laboratory fluorescence-XAS at energies above approximately 20 keV remains challenging. Silicon-based detectors progressively lose effectiveness in the hard-X-ray range, whereas high-purity germanium detectors offer excellent spectroscopic performance but require cryogenic cooling. Conversely, high-Z compound semiconductors such as CdTe and CdZnTe provide high

hard-X-ray absorption efficiency and room-temperature operation, although their use requires careful control of charge transport, detector segmentation, electronic noise and spectral reconstruction.

As a consequence, the K-edges of technologically relevant elements such as Mo, Rh, Ag and Cd remain difficult to investigate by laboratory fluorescence-XAS under realistic experimental conditions. There is therefore a clear need for a compact platform combining high-resolution crystal optics with efficient, room-temperature hard-X-ray detection. Addressing this technological gap would extend laboratory XAS beyond 20 keV and establish a complementary capability between conventional laboratory instruments, synchrotron facilities and future compact photon sources.

2 Goals and Expected Outcome (2000 caratteri)

The **main goal** of the project is **to develop and validate a room-temperature laboratory platform for fluorescence-mode X-ray absorption spectroscopy (f-XAS) above 20 keV**. Source, crystal optics, sample geometry, CZT detection, electronics and simulations will be optimized as a single measurement chain to obtain chemically meaningful XANES spectra from samples for which transmission is impractical.

The project is organized into the following specific **goals (G)** and **expected outcomes (EO)**:

G1 – Establish a validated design and predictive model of the complete instrument.

EO1 – An integrated simulation workflow and a frozen experimental design, validated at the Ag K-edge and at one energy of at least 40 keV. Predictions and measurements will agree within 20% for detected rate and within 10% for spot size and energy resolution.

G2 – Develop and characterize a multichannel CZT detector operating at room temperature.

EO2 – A calibrated system with intrinsic efficiency above 90% at 40 keV, energy resolution of 1 keV FWHM or better over 20–40 keV, dead time below 10% and residual pile-up below 5% at the selected operating rate.

G3 – Validate the complete platform on reference compounds with different chemical states.

EO3 – Quantitative Ag K-edge XANES with effective XAS resolution of 10 eV FWHM or better, edge-position reproducibility within 1 eV, edge-step SNR of at least 20 and acquisition time below the 12 h laboratory benchmark. The normalized spectral deviation from reference data will be below 5%.

G4 – Demonstrate f-XAS on at least one thick, diluted or operando-compatible sample.

EO4 – Discrimination of at least two chemical states through an edge shift of at least 1 eV and/or a normalized white-line variation of at least 5%, with a significance of at least 3σ and an acquisition time below 12 h per spectrum. The measured operating domain will be reported in terms of concentration, sample geometry and acquisition time.

3 Methodology (10000 caratteri)

PRISM will follow a requirements-driven and simulation-informed development strategy. The experimental requirements will first be translated into quantitative specifications for the X-ray source, crystal optics, sample geometry, CZT detector and DAQ. End-to-end simulations will then guide the design and co-optimization of these subsystems before procurement and construction. The resulting instrument will be assembled together with the custom multichannel CZT detection system and

validated through a staged programme, progressing from subsystem characterization and calibration to measurements on reference compounds and, finally, realistic samples. Experimental results will be continuously compared with the simulations, establishing a closed simulation–measurement loop that will support design refinement, performance verification and reliable interpretation of the measured spectra.

Requirements, simulations and design freeze

An end-to-end, source-to-detector simulation of the spectrometer will be developed to define the experimental requirements and freeze the main design parameters before procurement and construction. The Ag K-edge at approximately 25.5 keV will be used as the reference case for defining the XANES measurement protocol and the required performance in terms of energy resolution, photon flux, signal-to-background ratio and acquisition time [11].

The simulation workflow will consist of two coupled stages. First, SHADOW4 ray-tracing simulations [12] building on the approach previously validated for the VOXES spectrometer [13], will model photon transport from the X-ray tube through the collimating slits and crystal optics to the sample. Different silicon- and germanium-based crystal reflections will be investigated in the VOXES-type von Hamos geometry. Parametric studies will optimize the crystal material, reflection, curvature radius ρ_{cpc} , dimensions, source–crystal and crystal–sample distances, slit apertures and Bragg geometry. The simulations will quantify the geometrical acceptance, energy bandwidth, photon flux, beam profile and illuminated spot size at the sample, while also evaluating alignment and mechanical tolerances.

The SHADOW4 output will then be transferred to GEANT4, which will simulate photon absorption in the sample, fluorescence emission, self-absorption, elastic and inelastic scattering, background contributions and energy deposition in the multichannel CZT detector. The combined simulations will be used to co-optimize the source, crystal optics, slit system, sample geometry and detector configuration, balancing count rate, spectral resolution and background rejection.

A coordinated motorized scanning scheme will also be defined to vary the incident energy across the Ag K-edge while maintaining the required source–crystal–sample geometry. The final outcome of this activity will be a design freeze specifying the X-ray source, crystal type and dimensions, curvature radius, distances, slit apertures, sample position, CZT geometry, motion stages and required positioning accuracy.

CZT/DAQ development and instrument integration

To measure the fluorescence spectrum emitted by the sample at energies above 20 keV, a custom multichannel cadmium-zinc-telluride (CZT) detector system will be designed and implemented. Unlike general-purpose commercial detectors, its geometry, segmentation and number of channels will be defined from the photon-flux, background and count-rate requirements established in WP1. A baseline crystal thickness of 2 mm will provide high detection efficiency in the target energy range while maintaining compactness and room-temperature operation; the final thickness and active area will be confirmed through GEANT4 simulations. Sensitivity at lower energies will also be retained to identify and reject background components or contamination arising from lower-order crystal reflections.

Dedicated low-noise front-end electronics and a multichannel data-acquisition system will be developed to provide synchronized waveform acquisition, stable triggering and channel-by-channel calibration. Pulse-reconstruction and equalization algorithms will correct differences in gain, baseline and response among channels. Pulse-shape-analysis techniques, building on methods previously developed for germanium detectors, will be used to reject pile-up and distorted events and to mitigate the effects of incomplete charge collection and charge trapping in the CZT crystals. Dead-time and pile-up corrections will be implemented to preserve spectral accuracy over the expected range of incident rates.

The detector system will be characterized in terms of absolute and relative efficiency, energy resolution expressed as FWHM, gain stability, linearity and maximum useful count rate. These measurements will define the operating point that provides the best compromise between efficiency, spectral resolution and acquisition time. Finally, the CZT array, front-end electronics and DAQ will be integrated with the detector mechanics, motor-control system and instrument input/output monitoring, enabling coordinated acquisition with the monochromator energy scan and continuous recording of the relevant experimental parameters. This activity forms part of the broader requirements-driven development of the PRISM fluorescence-XAS instrument.

Commissioning and quantitative reference validation

Following assembly and integration, the instrument will undergo a structured commissioning and quantitative validation programme. The multichannel CZT detector will first be calibrated channel by channel to determine the energy scale, gain, offset, linearity and energy resolution. In parallel, the incident-energy scale of the monochromator will be established using reference absorption features and verified across the scan range. Dedicated measurements will characterize the beam spot at the sample, energy bandwidth, photon rate, spectral background and stability, providing a direct comparison with the SHADOW4 and GEANT4 predictions. The detector response will also be tested at energies extending to at least 40 keV to verify the performance of the system beyond the Ag K-edge.

The complete XAS measurement chain will then be validated at the Ag K-edge. Repeated XANES scans of a metallic Ag foil will be used to assess energy calibration, scan-to-scan reproducibility, signal-to-noise ratio and acquisition time. At least two additional Ag compounds representing different oxidation states or local chemical environments will subsequently be measured. Their edge positions and XANES features will be compared with reference spectra to verify that the instrument can distinguish chemically meaningful differences. The final validation will quantify the achieved energy resolution, background rejection, reproducibility, stability and useful count rate, and will determine whether the performance requirements defined during the design phase have been met.

Realistic-sample demonstration

After validation on reference materials, the capability of PRISM will be demonstrated on a realistic Ag-containing sample selected to represent conditions in which fluorescence detection is advantageous, such as a thick, diluted, heterogeneous or operando-compatible specimen. Two controlled and reproducible chemical states will be prepared to produce measurable differences in the Ag local electronic environment while keeping the sample composition and geometry sufficiently well characterized for quantitative comparison.

Before the measurements, GEANT4 simulations and analytical attenuation models will be used to estimate incident-beam absorption, matrix effects and self-absorption of the emitted fluorescence. Repeated fluorescence-XAS scans will then be acquired under identical conditions to evaluate reproducibility and sensitivity to chemical changes. The spectra will be analysed in terms of absorption-edge position, white-line intensity and other relevant XANES features, and the statistical significance of the observed differences will be quantified.

Measurements and simulations will finally be repeated for selected Ag concentrations, sample thicknesses, geometries and acquisition times. This study will define the practical operating limits of the instrument and establish the conditions under which reliable chemical-state discrimination can be achieved on realistic samples. The outcome will therefore be both a proof-of-capability measurement and a quantitative performance map linking sample properties and measurement geometry to signal-to-noise ratio, spectral distortion and required acquisition time.

4 Organisation (4000 caratteri)

The project will be organised as follows:

Name	Unit	FTE	Responsibility/WP
Simone Manti	INFN, Section of Frascati	1.0	Scientific Coordinator; overall coordination; WP1–WP4
Alessandro Scordo	INFN, Section of Frascati	0.3	Crystal optics, ray tracing, integration; WP1/WP3
Catalina Curceanu	INFN, Section of Frascati	0.1	Scientific review and application validation; WP4
Massimiliano Bazzi	INFN, Section of Frascati	0.1	Electronics/DAQ; WP2

Simone Manti will serve as Scientific Coordinator, subject to formal appointment by the Director of the host structure in agreement with CSN5, and will act as guarantor to the Committee for the proper progress of the project. He will coordinate the scientific programme, the interfaces among WPs, schedule and risk monitoring, and annual and final reporting.

PRISM is structured into four interdependent WPs, whose timeline is shown in Figure 4. WP1 (M1–M6) defines the quantitative requirements and uses SHADOW4 and GEANT4 to freeze the source–optics–sample–detector geometry. WP2 (M4–M15) develops and characterizes the multichannel CZT detector, low-noise front-end, DAQ and signal-processing chain. WP3 (M10–M18) covers procurement, mechanical assembly, subsystem integration and commissioning. WP4 (M16–M24) validates the instrument with Ag foil and reference compounds, demonstrates f-XAS on a realistic sample and determines the practical operating domain.

The planned overlap between the WPs enables early procurement and supports an iterative simulation–measurement loop. Monthly technical meetings and formal reviews at each milestone will be used to monitor schedule, technical performance and project risks.

WORK PACKAGE (WP)	PLANNED ACTIVITY (A)	Year I				Year II			
		M1-M3	M4-M6	M7-M9	M10-M12	M13-M15	M16-M18	M19-M21	M22-M24
WP1 - Requirements, simulations and design freeze	A1.1 - Requirements definition	■							
	A1.2 - SHADOW4 ray-tracing optimization		■						
	A1.3 - GEANT4 fluorescence simulations		■						
	A1.4 - Final geometry selection		■	◆					
WP2 - CZT, front-end, DAQ, calibration and characterization	A2.1 - CZT detector and front-end design		■	■	■				
	A2.2 - Multichannel DAQ and signal-processing development			■	■	■			
	A2.3 - CZT calibration and performance characterization				■	■	■		
	A2.4 - Detector validation and operating-point selection					■	◆		
WP3 - Procurement, mechanics, integration and commissioning	A3.1 - Procurement of source, optics and motion stages				■	■	■		
	A3.2 - Mechanical design and spectrometer assembly					■	■	■	
	A3.3 - Hardware, software and control-system integration					■	■	■	
	A3.4 - Instrument commissioning and verification						■	◆	
WP4 - Ag reference samples and realistic sample demonstration	A4.1 - Ag foil calibration and reproducibility tests						■	■	
	A4.2 - XANES validation on Ag reference compounds						■	■	■
	A4.3 - f-XAS demonstration on a realistic sample							■	■
	A4.4 - Performance map and operating-domain assessment							■	◆

Figure 4. Gantt table of the PRISM objected divided in each Work Packages (WPs) and for each one its planned activity (A) thru the two years and milestone are indicated by the yellow mark.

The project **milestones** are:

- **MS1 – Instrument design frozen (M6).** Experimental requirements, complete source–optics–sample–detector geometry and procurement specifications approved.
- **MS2 – CZT detection chain validated (M15).** Intrinsic efficiency above 90% at 40 keV, energy resolution of 1 keV FWHM or better, dead time below 10% and residual pile-up below 5% at the selected operating rate.
- **MS3 – Integrated instrument commissioned (M18).** Synchronized scanning and energy calibration operational; measured rate within 20%, and spot size and energy resolution within 10% of simulation predictions.
- **MS4 – f-XAS capability demonstrated (M24).** Ag K-edge XANES obtained with SNR ≥ 20 and edge-position reproducibility within 1 eV; two chemical states discriminated in a realistic sample with a significance of at least 3σ ; operating domain documented.

The project **deliverables** (D) are:

- **D1** – Requirements matrix and Ag K-edge measurement protocol (M3).
- **D2** – Validated SHADOW4–GEANT4 workflow, frozen instrument design and procurement specifications (M6).
- **D3** – CZT calibration and performance-characterization dataset with acceptance report (M15).
- **D4** – Ag foil and reference-compound XANES dataset with quantitative validation report (M21).
- **D5** – Realistic-sample dataset, operating-domain assessment, documented analysis tools and final project report (M24).

5 Synergies with Other Activities (3000 caratteri)

PRISM is embedded in complementary INFN activities while retaining independent objectives, budget and deliverables.

At LNF, PRISM builds directly on VOXES, developed through an INFN-CSN5 Young Researchers Grant. It will reuse established know-how and, where available, instrumentation for bent-crystal

optics, alignment, mechanics, X-ray-source operation and spectral analysis. The LNF Directorate has committed access to laboratory resources and instrumentation. This institutional contribution is in kind; no separate cash co-funding is included.

Cross-committee synergy arises from SIDDHARTA-2/KAONNIS (CSN3), through shared expertise in precision X-ray detection, semiconductor-detector characterization, low-noise readout, calibration, dead-time and pile-up control, and spectral reconstruction. This expertise will support the PRISM CZT system, while PRISM will return methods relevant to hard-X-ray detector characterization. No CSN3 financial contribution is assumed.

PRISM is also strategically complementary to EuAPS, the INFN-led PNRR project involving CNR and the University of Rome Tor Vergata and developing a betatron X-ray user facility at SPARC_LAB. After laboratory validation, PRISM will use its measured source and detector requirements to assess transferability to this compact pulsed source and, if operating conditions permit, define a pilot test. EuAPS funding and resources are not included in the PRISM budget, and EuAPS availability is not required to achieve the project deliverables.

Rigaku and Quantum Design have been approached regarding the scientific and technological relevance of laboratory fluorescence-XAS above 20 keV. If the letters of support are signed, their role will be limited to following the project and exploring future technical exchanges and applications. No industrial cash or in-kind co-funding is currently committed, and any procurement will follow INFN procedures without preferential treatment.

6 Impact and Spin-offs (3000 caratteri)

PRISM will establish within INFN a new laboratory capability for quantitative fluorescence-XAS above 20 keV using room-temperature detection, a regime where efficient non-cryogenic spectroscopy remains challenging. Its impact will extend beyond one instrument: integrating crystal optics, multichannel CZT, low-noise electronics and DAQ with SHADOW4-GEANT4 modelling will consolidate at LNF expertise in semiconductor detectors, precision X-ray spectroscopy, radiation transport and spectral reconstruction. The platform, validated models, calibration protocols and performance maps will be reusable by INFN activities developing hard-X-ray instrumentation. PRISM will valorize existing VOXES infrastructure and strengthen INFN's capacity to translate detector R&D from simulation and subsystem tests into validated instruments. After laboratory validation, its design rules may also inform EuAPS. The project will provide interdisciplinary training for young researchers and technical staff.

For the wider scientific community, PRISM addresses the limited availability of laboratory fluorescence-XAS at the K-edges of technologically relevant elements such as Mo, Rh, Ag and Cd. Meeting its quantitative targets will enable routine screening, method development, sample and cell optimization and long in-situ or operando studies on thick, dilute and heterogeneous specimens that are difficult to measure in transmission. This capability will complement, not replace, synchrotrons and XFELs: preliminary measurements and protocol optimization can make scarce beamtime more effective, while repeated scans and slowly evolving processes can remain in the laboratory. Beneficiaries span catalysis, electrochemistry, energy materials and advanced alloys. Documented simulation workflows, calibration methods, reference datasets and operating-domain maps will promote reproducibility and guide other laboratories in assessing and adapting the approach.

Potential technology-transfer outcomes include the modular room-temperature CZT detection chain—front-end electronics, multichannel DAQ, calibration and pulse-reconstruction methods—the integrated digital model of source, optics, sample and detector, and the compact hard-X-ray

spectrometer architecture. Following experimental validation, these assets could be assessed with INFN technology-transfer structures for protectable know-how, licensing or industrial co-development. Applications may include routine materials characterization, quality control and long-duration monitoring of catalysts, batteries and advanced materials. Technical exchange with X-ray-instrumentation companies, component suppliers and simulation-oriented INFN spin-offs could accelerate uptake. Industrial adoption will remain conditional on demonstrated performance, robustness, manufacturability, cost and market need.

7 Costs

The financial request is structured according to the technical development of the project and distributed over the two years of activity. During Year I, the budget is mainly devoted to the development of the fluorescence detection system, including the multichannel CZT detector, the digitizer and the bent Ge/Si crystal optics required for the selected absorption edges. These components are essential for the design, integration and initial commissioning activities carried out within WP2. The Year II request supports completion of the experimental platform through the acquisition of the X-ray tube and its power supply, motorized translation and rotation stages, slits, custom mechanical supports for the sample and detector, and reference targets for the validation campaign. Travel funds are reserved for experimental activities, collaboration meetings and dissemination of the results at an international conference, subject to acceptance of an oral contribution. All equipment items and all individual expenses exceeding €2,000 will be supported by detailed supplier quotations. Costs below this threshold are based on preliminary supplier or internal estimates. The requested amounts remain below the maximum annual funding limit and are directly linked to the activities and deliverables described in the corresponding Work Packages.

Cost category	Item / description	Year I (€)	Year II (€)	Total (€)
Equipment	Multichannel CZT detector	30000	0	30000
Equipment	Digitizer	20000	0	20000
Equipment	Bent crystal optics	15000	15000	30000
Consumables	Reference samples and targets	0	2000	2000
Equipment	Precision slits	0	2000	2000
Services	Custom sample and CZT supports	0	3000	3000
Equipment	Motorized translation stages	0	5000	5000
Equipment	Motorized rotation stages	0	3000	3000
Equipment	X-ray tube	0	7000	7000
Equipment	X-ray tube power supply	0	10000	10000
Equipment	Closed-loop cooling system	0	5000	5000
Consumables / services	Consumables and precision machining	0	4000	4000
Travel	Experimental activity at collaborating facilities	0	2000	2000
Travel	Project and collaboration meetings	0	1500	1500
Travel	EXRS 2028 conference (oral presentation only)	0	3000	3000
Total requested		65000	62500	127500

8 Risk Assessment

X-ray source, crystal optics and motorized stages

- Risk: Delivery or installation delays, or components failing interface or performance specifications.

- Consequence: Postponed assembly and commissioning, with less time for WP4 and potentially reduced flux, resolution or scan range.
- Mitigation: Define interfaces and acceptance criteria during WP1; order long-lead components immediately after the design freeze; identify compatible alternatives; perform incoming acceptance tests; use available VOXES components for preliminary tests where compatible.

CZT detector, front-end electronics and multichannel DAQ

- Risk: Component delays, failed channels, excessive noise, charge trapping, non-uniform response, dead time or pile-up.
- Consequence: Delayed detector validation and degraded useful count rate or fluorescence/background discrimination.
- Mitigation: Test the system progressively at channel and module level; optimize bias, grounding, shielding, shaping and operating rate; apply channel equalization, pulse-shape analysis and dead-time corrections. A validated sub-array may support early integration without relaxing the final acceptance criteria.

VOXES/LNF infrastructure and technical services

- Risk: Temporary unavailability or malfunction of the experimental area, X-ray source, shielding, interlocks, workshop or technical services.
- Consequence: Interruption of assembly, calibration or data taking.
- Mitigation: Schedule access and safety reviews in advance; perform preventive checks; include schedule contingency and staged bench tests. During downtime, continue simulations, electronics tests, control-software development, mechanical preparation and data analysis.

Crystal-optics and scanning system

- Risk: Crystal performance differs from predictions, or positioning repeatability, backlash or synchronization is inadequate.
- Consequence: Incident-energy resolution, calibration or scan reproducibility falls below the XANES requirements.
- Mitigation: Perform tolerance studies before procurement; define supplier metrology and bench-acceptance tests; retain adjustable mounts, slits and distances and an alternative crystal/reflection option; recalibrate the energy-position relation using reference absorption features.

End-to-end model and integrated performance

- Risk: Measured flux, background, self-absorption or CZT response differs from SHADOW4–GEANT4 predictions, resulting in insufficient SNR or excessive acquisition time.
- Consequence: A suboptimal design may be frozen, or G3/G4 may not be achieved in the initial configuration.
- Mitigation: Propagate uncertainties and validate each simulation stage against subsystem measurements; identify experimental losses and background sources; optimize source operation, slits, geometry, detector solid angle, shielding and reconstruction. Complete Ag-reference validation before proceeding to WP4.

Ag references, realistic sample and sample environment

- Risk: Reference materials are delayed, or the selected sample is unstable, insufficiently reproducible or affected by excessive attenuation, self-absorption or contamination.
- Consequence: Delayed WP4 measurements or ambiguous chemical-state discrimination.

- Mitigation: Procure and characterize Ag standards early; preselect alternative concentrations, thicknesses and geometries. If an operando-compatible environment is unavailable, use a controlled thick, diluted or heterogeneous static sample preserving the same chemical observable.

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